HACK THE COAST 2026

AI Product Discovery & Trend Intelligence Tool

Problem Background

Prince of Peace (PoP) is a CPG distributor with ~800 products across health foods, herbal teas, ginseng, ginger chews, and Tiger Balm. The company sources products globally and sells to 100,000+ retail outlets across the U.S.

Today, PoP's buyers find new products manually: attending trade shows, scanning social media, and relying on personal networks. There is no systematic way to scan for emerging product trends, rising ingredients, or new market entrants. By the time a trend is identified and a compliant supplier is found, competitors have often already filled the shelf.

In this challenge, students will build a product discovery and trend intelligence tool that:

- Scans public data sources for emerging food, beverage, and wellness trends
- Filters opportunities against PoP's specific sourcing criteria
- Surfaces actionable recommendations for both distribution and product development
- Is usable by non-technical buyers

No prior CPG industry experience is required.

Problem Statement

Design and build a product discovery tool that helps PoP's buying team identify emerging products, ingredient trends, and category shifts before competitors, using publicly available data.

The tool should go beyond generic trend reporting. It should filter against real business constraints and surface opportunities that are actionable.

What the System Must Do (Requirements)

Core Discovery

- Ingest data from at least 2 public sources (e.g., Amazon, Google Trends, social media, FDA databases, trade publications)
- Identify trending products, ingredients, or categories in food, beverage, wellness, or personal care
- Rank or score opportunities by signal strength

Business Filtering

- Filter against PoP's sourcing criteria:
  - Shelf life must exceed 12 months
  - Ingredients must not appear on FDA restriction lists
  - Flag products from countries with high tariff or trade restriction risk
- Categorize opportunities by PoP's product categories (dry goods, confections, teas, personal care, health & wellness, etc.)

Product Development Angle

- Identify ingredient trends that PoP could apply to its existing product lines (ginger, ginseng)
- Example: if kombucha is trending, suggest a kombucha-ginger product. If adaptogens are trending, suggest a ginseng-adaptogen functional snack.
- Flag opportunities for both "distribute an existing product" and "develop a new PoP-branded product"

Output & Usability

- Present results in a format a non-technical buyer can understand and act on
- Include enough context per recommendation for a buyer to decide whether to investigate further

---

Suggested Approach

Instead of trying to build a complete platform, focus on a clear pipeline:

- **Data Collection:** Pick 2-3 public data sources and pull relevant signals (trending search terms, bestseller movement, new product listings, social media mentions)
- **Scoring & Ranking:** Define a simple scoring model. What makes a trend signal strong vs. weak? Recency, growth rate, category relevance, and competition level are good starting factors.
- **Filtering:** Apply PoP's sourcing constraints (shelf life, FDA, tariff) to eliminate non-viable opportunities before they reach the buyer.

- **Presentation:** Build a simple interface or report that shows top recommendations with context: what the trend is, why it's relevant to PoP, and what action the buyer should consider.

A focused prototype covering one product category well is better than a broad but shallow scan across everything.

Team Collaboration Model

Teams should divide responsibilities, for example:

Data & Pipeline

- Data source selection and scraping/API integration
- Data cleaning and normalization
- Scoring model design

Business Logic & Filtering

- Sourcing criteria filter implementation
- Category mapping and taxonomy
- Product development opportunity flagging

Frontend & Presentation

- Dashboard or report design
- User experience for non-technical buyers
- Sample output generation

Suggested 2-Day Plan

Day 1

- Select data sources and build data collection pipeline
- Define scoring model criteria
- Implement sourcing criteria filters (shelf life, FDA, tariff)
- Start building presentation layer

Day 2

- Refine scoring and ranking logic
- Generate sample recommendations for 1-2 product categories

- Build demo-ready output with real examples
- Test with PoP's product catalog for relevance
- Prepare presentation

Deliverables

Each team should present:

1. How the tool discovers trends (data sources, pipeline, scoring logic)
2. How it filters against PoP's sourcing criteria
3. Sample output: top recommendations for at least one product category, with context for each
4. How it identifies product development opportunities (not just distribution)
5. Key challenges and lessons learned

Presentation time: 6-8 minutes